Experiment No -03 (Group A)

Title:-

Search Techniques in an E-commerce System Using C programing

Problem statement:-

In an e-commerce system, customer account IDs are stored in a list, and you are tasked with writing a program that implements the following:

- Linear Search: Check if a particular customer account ID exists in the list.
- Binary Search: Implement Binary search to find if a customer account ID exists, improving the search efficiency over the basic linear

Pre-requisite:-

- Basic understanding of C programming.
- Knowledge of variables, array, loops, and functions.
- Understanding of conditional statements (if, else).
- Basic knowledge of searching algorithms.

Objective:-

- To understand the concept of searching techniques.
- To implement **Linear Search** for finding a customer account ID in a list.
- To implement **Binary Search** to improve search efficiency on a sorted list.
- To compare the performance of Linear Search and Binary Search.

Software and Hardware requirements:-

1. **Operating system:** Linux- Ubuntu 16.04 to 17.10, or Windows 7 to 10,
2. **RAM-** 2GB RAM (4GB preferable)
3. IDE :- Turbo C

Outcome:-

After completing this experiment, the student will be able to: Implement Linear Search and Binary Search in Python. Search for a customer account ID in a list. Understand the difference between sequential and divide-and-conquer searching techniques. Compare the efficiency of Linear Search and Binary Search. Identify that Binary Search performs better on large, sorted datasets.

Theory:-

Sequential Search/Linear Search

Sequential Search/Linear Search is the technique in which given list of elements is scanned from the beginning.

The key element is compared with every element of the list. If the match is found the searching is stopped otherwise it will be continued to the end of the list.

Although this is a simple method there is some unnecessary comparison involved in this method.

The time complexity of this algorithm is O(n). The time complexity will increase linearly with value of n.

Sequential Search/Linear Search Algorithm-

1. Start.
2. Read the array arr[], the number of elements n, and the search key as key.
3. Set index i = 0.
4. Repeat while i < n:
5. Compare if arr[i] == key:

   Display **"Customer Account ID found at position i."**
6. Otherwise, increment i by 1.
7. If the loop completes without finding the key , Display **"Customer Account ID not found."**
8. Stop.

Advantages of Sequential Search/Linear Search-

1. It is simple to implement
2. It does not required specific ordering before applying the method.

Disadvantages of Sequential Search/Linear Search-

1. It is less efficient.

Real-life application of Linear Search.

1. Searching for a contact in an unsorted contact list.
2. Finding a product in a small inventory.

Binary Search :-

Binary Search is a searching algorithm in which the list of elements is divided into two sub-list & the key element is compared with the middle element.

If the match is found then , the location of the middle element is returned otherwise we search into either the halves (sub-list) depending upon the result produced through match.

Algorithm Binary Search

1. Start.
2. Read the sorted array arr[], lower index low, upper index high, and the search key key.
3. Repeat while low <= high:

    Calculate the middle index: mid = (low + high) / 2

    Compare if arr[mid] == key:

        Return mid (element found).

    else if arr[mid] < key:

        Set low = mid + 1 (search in the right half).

    else:

        Set high = mid - 1 (search in the left half).
4. If low > high, the element is not present in the array.
5. Return -1.
6. Stop.

Advantages of Binary Search-

It is efficient technique

Disadvantages of Binary Search

1) It requires specific ordering before applying the method

2) It is complex to implement

Real-life application of Binary Search.

1. Searching products in a sorted e-commerce database.
2. Finding a word in a dictionary.
3. Searching names in a sorted employee database.

Conclusion:-

The experiment successfully demonstrates the implementation of **Linear Search** and **Binary Search** for locating customer account IDs in a list.

Viva Questions:-

1. What is Linear Search?
2. What is Binary Search?
3. What is the prerequisite for Binary Search?
4. Which search algorithm works on unsorted data?
5. Why is Binary Search faster than Linear Search?
6. What is the time complexity of Linear Search?
7. What is the time complexity of Binary Search?
8. What is the space complexity of Linear Search?
9. What is the space complexity of Binary Search (Iterative)?
10. What is the middle element in Binary Search?
11. Give a real-life application of Linear Search.
12. Give a real-life application of Binary Search.